

Name _____

Date April 24, 2025

Total
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/16

1. Differentiate the following functions. Fully factor and simplify.

a) $f(x) = 3^{x^2+3x} + 2e$

$f(x) = 3^{x^2+3x} + 2e$

$f'(x) = 3^{x^2+3x} (\ln 3)(2x+3) + 2e$

b) $f(x) = \frac{e^{5t} + e^{6t}}{e^{3t}}$

$f(x) = \frac{e^{5t} + e^{6t}}{e^{3t}} = (3e^{3t})(e^{5t} + e^{6t})$

$f'(x) = \frac{(5e^{5t} + 6e^{6t})(e^{3t}) - (e^{5t} + e^{6t})(3e^{3t})}{(e^{3t})^2}$

$f'(x) = \frac{(5e^{5t} + 6e^{6t}) - 3(e^{5t} + e^{6t})}{e^{3t}}$

$f'(x) = \frac{(2e^{5t} + 3e^{6t})}{e^{3t}}$

2. Determine the equation of the tangent to $f(x) = e^{-(x-2)^2}$ at $x = 1$. Use exact values.

$f(x) = e^{-(x-2)^2}$

$f(x) = e^{-(x^2 - 4x + 4)}$

$f(x) = e^{-x^2 + 4x - 4}$

$f'(x) = e^{-x^2 + 4x - 4} (-2x + 4)$

$f'(1) = e^{-(1)^2 + 4(1) - 4} (-2(1) + 4)$
 $= e^{-1} (2)$
 $= 2e^{-1} \leftarrow m$

$P(1, e^{-1})$

$y = mx + b$

$e^{-1} = 2e^{-1}(1) + b$

$b = e^{-1} - 2e^{-1}$

$f'(x) = e^{3t} (2e^{2t} + 3e^{3t})$

$f'(x) = 2e^{2t} + 3e^{3t}$

$b = -e^{-1}$

$y = 2e^{-1}x - e^{-1}$

$y = \frac{2x}{e} - \frac{1}{e}$

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1. 3. Find the absolute extrema of $g(x) = x^2 e^{-2x^2 - 2x}$ on $[-1, 1]$. Provide exact points.

$$g(x) = x^2 e^{-2x^2 - 2x}$$

$$g'(x) = 2x(e^{-2x^2 - 2x}) + x^2(e^{-2x^2 - 2x})(-4x - 2)$$

$$= (e^{-2x^2 - 2x}) [2x + x^2(-4x - 2)]$$

$$= (e^{-2x^2 - 2x}) (2x - 4x^3 - 2x^2)$$

$$= (e^{-2x^2 - 2x}) (x) (-4x^2 - 2x + 2)$$

$$C = 2x(e^{-2x^2 - 2x})(-2x^2 - x + 1)$$

$$\downarrow$$

 $x = 0$

no solutions

$$\downarrow$$

 $x = \frac{1}{2}, -1$

$f(-1) = 1$

$f(1) = \frac{1}{e^4} \rightarrow 0.018$

$f(0) = 0$

$f\left(\frac{1}{2}\right) = \frac{1}{4}e^{-1.5} \rightarrow 0.0557$

the absolute
min occurs @
 $P(0, 0)$ and
the absolute
max occurs
@ $P(-1, 1)$

4. The effectiveness of studying for an exam depends on how many hours a student studies. Some experiments show that if the effectiveness, E , is put on a scale of 0 to 10, then $E(t) = 0.5 \left[10 + te^{-\frac{t}{20}} \right]$, where t is the number of hours spent studying for an examination. If a student has up to 30 hours for studying, how many hours are needed for maximum effectiveness?

$$E(t) = 0.5 \left[10 + te^{-\frac{t}{20}} \right]$$

$$E'(t) = 0.5 \left[(1)(e^{-\frac{t}{20}}) + (t)(e^{-\frac{t}{20}})\left(-\frac{1}{20}\right) \right]$$

$$0 = (0.5)(e^{-\frac{t}{20}}) \left[1 - \frac{t}{20} \right]$$

no solutions

$$\downarrow$$

 $1 - \frac{t}{20} = 0$

$$\frac{t}{20} = 1$$

$$t = 20$$

$E(0) = 5$

$E(20) = 8.6788$

$E(30) = 8.3469$

studying for 20h
results in max
efficiency @ a point
at 8.6788

$P(20, 8.6788)$